

Student Loan Application Funnel Optimizer

Analytics case study — funnel diagnosis and onboarding A/B test

Problem

Student loan applicants pass through five stages — acquisition, eligibility, onboarding, document submission, and approval — and drop off at each one. The business needs to know where drop-off is worst, whether it varies by applicant segment, and whether a proposed onboarding redesign is actually worth shipping.

Approach

Built a 20,000-row simulated application dataset with realistic, non-uniform drop-off probabilities (credit tier genuinely drives eligibility and approval odds, rather than uniform random chance). Computed stage-wise conversion and drop-off, segmented outcomes by credit tier and acquisition channel, and ran a controlled A/B test on a simplified onboarding flow using a two-proportion z-test with 95% confidence intervals.

Key Findings

- **25.7% end-to-end approval rate** (5,147 of 20,000). Drop-off is spread fairly evenly across all four transitions (27–32% lost at each stage) — there is no single dominant bottleneck.
- **Simplified onboarding produced a statistically significant +8.1 percentage-point lift** in completion: 69.4% (control) → 77.6% (treatment), $z = 10.97$, $p < 0.001$.
- **Credit tier drives eligibility far more than acquisition channel drives approval**: eligibility pass rate ranges from 55.5% (Thin File) to 89.5% (Excellent), while approval rate across channels varies by only 0.9 points (25.4–26.3%).

Recommendation

Ship the simplified onboarding flow — the lift is real and statistically defensible, not a cosmetic improvement. Because drop-off is distributed across every stage rather than concentrated in one, the same test-and-measure approach used here for onboarding should be extended to the document-submission and approval stages next, rather than assuming one fix will resolve the whole funnel. Acquisition-channel optimisation is a lower-priority lever, since channel has minimal effect on end-to-end approval.

KPIs Tracked

Stage-wise conversion rate | Drop-off rate per transition | Onboarding completion rate (by A/B group) | Eligibility pass rate (by credit tier) | End-to-end approval rate (by acquisition channel)

Full analysis code, dataset, and interactive dashboard: see project repository.